Delayed Mode Selection and Parity-Separated Observables in a Rapidly Rotating Fluid Column with Event-Triggered Reinjection

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Abstract

We study a classical rotating-fluid platform in which axisymmetric forcing, finite axial transport time, and event-triggered reinjection combine to form a delayed nonlinear loop. The starting point is a rapidly rotating cylindrical column driven by a bottom rotor under a steady background rotation. In the low-Rossby, low-Ekman regime, axisymmetric vertical forcing generates a delayed axial response governed by rotating-fluid transport rather than instantaneous communication. We show how this delayed transport can be converted into a regenerative feedback loop by triggering each subsequent rotor pulse from a detected surface event after a prescribed digital latency.

Two distinct observable sectors are then naturally separated by parity. The first is an odd sector, represented by signed azimuthal or angle-like response variables, which reverses under reversal of the axial forcing sign. The second is an even sector, represented by scalar response measures quadratic in velocity or swirl amplitude, which remain invariant under sign reversal. This parity split implies that a symmetric up/down forcing protocol can exhibit leading-order cancellation in the odd sector while retaining a nonzero accumulated contribution in the even sector.

To connect the fluid platform to delayed nonlinear dynamics, we formulate a reduced event-triggered loop model with effective delay τ_{eff} and gain κ_{eff}. In the reinjected regime, the platform becomes a classical delayed-feedback system admitting mode selection, branch survival, multistability, and transition to irregular dynamics as the effective gain-delay product is increased. We outline experimentally measurable criteria distinguishing one-pass delayed transport from genuine delayed-loop branch selection. The results require no quantum assumptions and identify a rotating-fluid route from finite propagation time to classical order, hysteresis, and chaos.

1 Introduction

Closed dynamical loops with finite propagation speed form a broad class of classical systems in which delay is not merely a correction to local dynamics, but a constitutive ingredient. In optical rings, electronic delay lines, and phase-locked feedback architectures, finite transit time can organize the dynamics into stable operating branches, multistable regimes, or irregular motion. Separately, rapidly rotating fluids support strongly anisotropic transport and wave propagation, with axial communication governed by rotation-induced structure rather than by the kinematics of non-rotating flow.

The present work combines these two themes in a single classical platform: a rapidly rotating cylindrical fluid column with axisymmetric forcing and event-triggered reinjection. The physical picture is simple. A bottom rotor, embedded in a slowly rotating cylinder, launches a disturbance into a rotation-prepared medium. Under suitable conditions the response is not dominated by

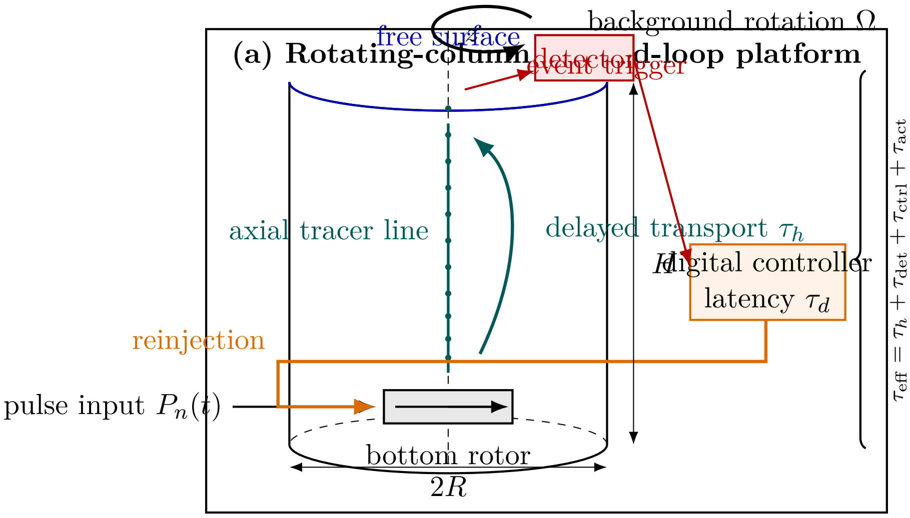

Figure 1: (a) A bottom rotor injects pulses into a rapidly rotating fluid column, generating a delayed axial response detected at the free surface; the detected event is digitally reinjected, closing the loop with effective delay τ_{eff}.

immediate radial spreading, but by delayed axial transport toward the free surface. Once that delayed response is used to trigger a subsequent rotor pulse, the platform is promoted from a one-pass transport experiment to a regenerative delayed loop.

This permits two conceptually distinct questions to be addressed within one setup. The first is a rotating-fluid question: under what conditions does axisymmetric forcing generate a reproducible delayed axial response rather than a predominantly divergent radial disturbance? The second is a nonlinear-dynamical question: once the detected arrival event is fed back to the actuator, does the resulting loop display branch selection, multistability, or transitions to irregular dynamics as a function of effective delay and gain?

A second theme of the present paper is parity separation. In the same rotating-fluid platform, different observables transform differently under reversal of the sign of the axial forcing. Signed angle-like observables are odd under reversal, while scalar quadratic measures built from speed, swirl intensity, or response amplitude are even. This creates a structurally sharp possibility: a symmetric forcing protocol may exhibit cancellation in one observable sector while accumulating a nonzero signal in another. The point is classical and kinematic. No microscopic interpretation is required for the parity statement itself.

The goal of this paper is deliberately narrow. We do not claim a new constitutive law for matter, nor any modification of standard wave or fluid theory. Instead, we isolate a minimal classical mechanism,

$$\text{rotation} + \text{finite transport time} + \text{event-triggered reinjection} \implies \text{delayed nonlinear loop dynamics}, \tag{1}$$

and identify a parity split,

$$\text{odd signed sector} \oplus \text{even scalar sector}, \tag{2}$$

within which delayed transport and reinjection create a platform for order, branch selection, hysteresis, and possibly chaos.

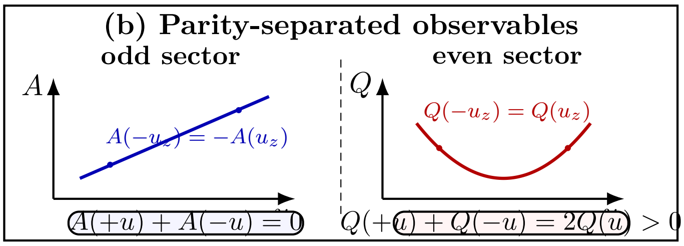

Figure 2: (b) Observable sectors separate by parity under reversal of the signed axial response variable u_z: odd signed observables cancel under reversal-symmetric protocols, whereas even scalar observables survive.

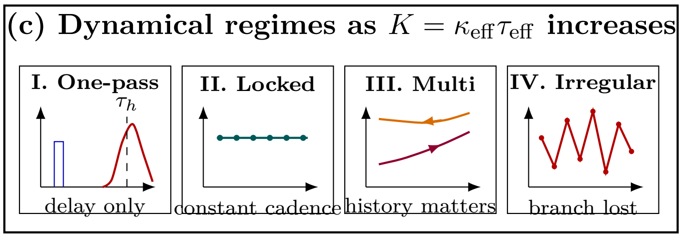

Figure 3: (c) As the effective gain-delay product $K = \kappa_{\text{eff}}\tau_{\text{eff}}$ increases, the platform is expected to pass through one-pass delayed transport, stable locking, multistability, and irregular-response regimes.

2 Governing framework

2.1 Rotating-column setting

Consider an incompressible fluid of density ρ and kinematic viscosity ν contained in a vertical cylinder of radius R and height H. The cylinder rotates with constant angular velocity

$$\mathbf{\Omega} = \Omega\,\hat{\mathbf{z}}, \tag{3}$$

and a compact rotor located near the bottom center provides short, approximately axisymmetric forcing pulses superposed on the slowly rotating base state.

In the rotating frame, the velocity field $\mathbf{u}(\mathbf{x}, t)$ and pressure field p satisfy

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla)\mathbf{u} + 2\mathbf{\Omega} \times \mathbf{u} = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f}, \tag{4}$$

together with incompressibility,

$$\nabla \cdot \mathbf{u} = 0. \tag{5}$$

Here $\mathbf{f}(\mathbf{x}, t)$ denotes the forcing generated by the rotor pulses.

We are interested in the regime

$$\mathrm{Ro} = \frac{U}{2\Omega L} \ll 1, \qquad \mathrm{Ek} = \frac{\nu}{\Omega L^2} \ll 1, \tag{6}$$

where U and L are characteristic induced speed and length scales. In this regime, the background rotation strongly organizes the response, and the dominant transport is anisotropic rather than isotropically divergent.

2.2 Axisymmetric vertical forcing and vorticity production

For axisymmetric forcing, the vertical vorticity

$$\omega_z = (\nabla \times \mathbf{u}) \cdot \hat{\mathbf{z}} \tag{7}$$

is coupled to the vertical velocity $w = \mathbf{u} \cdot \hat{\mathbf{z}}$ through the rotating-fluid dynamics. To leading order, one obtains the rotating-fluid production relation

$$\partial_t \omega_z = 2\Omega\, \partial_z w + \mathcal{R}, \tag{8}$$

where $\mathcal{R}$ collects nonlinear, viscous, and higher-order corrections. Equation (8) shows that vertical forcing in a rotating medium naturally produces opposite-sign vorticity above and below the forcing region.

This is the first structural point of the experiment: the column is not merely a container for pulses, but a medium whose background rotation transforms vertical forcing into a structured rotational response.

2.3 Finite axial transport time

The second structural point is that the response is not instantaneous. When the perturbation projects onto inertial-wave-like modes or other rotation-guided axial response channels, the surface signal appears only after a finite arrival time. In a reduced description, we denote that delay by

$$\tau_h. \tag{9}$$

For an idealized inertial-wave packet with wavevector $\mathbf{k} = (k_r, 0, k_z)$ and magnitude $k = \sqrt{k_r^2 + k_z^2}$, the rotating-fluid dispersion relation is

$$\omega = 2\Omega \frac{k_z}{k}, \tag{10}$$

and the axial group velocity is

$$c_{g,z} = \frac{\partial \omega}{\partial k_z}. \tag{11}$$

Hence the simplest arrival estimate is

$$t_{\mathrm{arr}} \sim \frac{H}{c_{g,z}}. \tag{12}$$

For the present paper, the exact microscopic transport channel is less important than the experimentally accessible fact that a reproducible finite delay exists between bottom forcing and surface detection. The measured delay τ_h is therefore treated as a primary effective parameter of the platform.

3 Event-defined reinjection loop

3.1 From one-pass transport to closed-loop dynamics

A delayed arrival by itself is only a transport phenomenon. To obtain a genuine delayed nonlinear loop, the detected arrival must be reinjected into the actuation.

Let t_n^{fire} denote the time at which the nth rotor pulse is applied, and let $s(t)$ denote a measured surface observable used for detection. This could be a local surface displacement, optical tracer intensity, pressure fluctuation, or a signed vertical-velocity proxy near the axis.

We define the nth detected arrival time by thresholding:

$$t_n^{\mathrm{det}} = \inf\left\{t > t_{n-1}^{\mathrm{fire}} + t_{\mathrm{ref}} : s(t) \geq s_{\mathrm{th}}\right\}, \tag{13}$$

where s_{th} is the detection threshold and t_{ref} is a refractory interval that suppresses multiple triggers from a single arrival or from ringdown.

Each detected event then schedules the next actuation after a fixed digital latency τ_d:

$$t_{n+1}^{\mathrm{fire}} = t_n^{\mathrm{det}} + \tau_d. \tag{14}$$

The resulting effective loop delay is

$$\tau_{\mathrm{eff}} = \tau_h + \tau_{\mathrm{det}} + \tau_{\mathrm{ctrl}} + \tau_{\mathrm{act}}, \tag{15}$$

where the separate terms denote hydrodynamic transport, detector latency, controller latency, and actuation latency.

3.2 Pulse protocol

A minimal rotor pulse may be written as

$$P_n(t) = A_n\, \Pi\left(\frac{t - t_n^{\mathrm{fire}}}{\Delta t_p}\right), \tag{16}$$

where A_n is the pulse amplitude, Δt_p is the pulse width, and Π is a rectangular window. More general pulse shapes may be used in practice, but (16) is sufficient for reduced modeling.

The simplest feedback prescriptions are

$$A_n = A_0 \qquad \text{(fixed-amplitude reinjection)}, \tag{17}$$

$$A_{n+1} = G\, A_n \qquad \text{(gain-controlled reinjection)}, \tag{18}$$

with G an effective feedback gain.

3.3 Reduced delayed-loop model

At the fully resolved hydrodynamic level, the reinjected platform is governed by the rotating Navier–Stokes equations together with the event-triggering logic. For analysis, however, it is useful to reduce the loop to an effective phase or response-amplitude description.

Let $\phi(t)$ denote a phase-like variable describing the state of the reinforced axial response. The minimal delay model is then

$$\dot{\phi}(t) = \omega_0 + \kappa_{\mathrm{eff}} \sin\big[\phi(t - \tau_{\mathrm{eff}}) - \phi(t)\big], \tag{19}$$

where ω_0 is an intrinsic response frequency and κ_{eff} is an effective loop coupling.

Equation (19) is not presented as an exact derivation from Eqs. (4)–(5). Rather, it is the minimal reduced delayed-loop normal form consistent with event-triggered reinjection and finite transport time. Its role is to organize the experimentally observed branches and their stability.

Uniformly rotating solutions

$$\phi(t) = \Omega_L t + \phi_0 \tag{20}$$

satisfy the locking condition

$$\Omega_L = \omega_0 - \kappa_{\text{eff}} \sin(\Omega_L \tau_{\text{eff}}). \tag{21}$$

Hence finite delay alone can generate multiple candidate branches

$$\Omega_{L,n} \approx \frac{2\pi n}{\tau_{\text{eff}}} + \delta\Omega_n, \qquad n \in \mathbb{Z}, \tag{22}$$

provided the gain-delay product is sufficiently large.

4 Parity-separated observables

4.1 Odd sector

Let u_z denote a signed axial response coordinate or a signed effective forcing variable. An observable A belongs to the odd sector if

$$A(-u_z) = -A(u_z). \tag{23}$$

Examples include signed angular displacement, signed azimuthal tracer shift, or signed vorticity increment.

In particular, if $\Delta\theta$ denotes a signed angle-like response, then symmetry under reversal of the signed axial episode implies

$$\Delta\theta(+u_z) = -\Delta\theta(-u_z). \tag{24}$$

4.2 Even sector

An observable Q belongs to the even sector if

$$Q(-u_z) = Q(u_z). \tag{25}$$

Examples include scalar measures quadratic in response speed or swirl intensity, such as

$$Q \propto u_z^2, \qquad Q \propto \omega_z^2, \qquad Q \propto |X|^2, \tag{26}$$

or cycle-integrated versions thereof.

The point is elementary but powerful:

$$(-u_z)^2 = (u_z)^2. \tag{27}$$

So if one protocol samples both signs of the axial response symmetrically, the even sector is sign-insensitive even when the odd sector is not.

4.3 Paired-protocol consequence

Consider a paired forcing protocol generating equal-magnitude opposite-sign response episodes $+u$ and $-u$. Then for odd and even observables one has, respectively,

$$A(+u) + A(-u) = 0, \tag{28}$$

$$Q(+u) + Q(-u) = 2Q(u) \geq 0. \tag{29}$$

Thus a reversal-symmetric protocol can exhibit cancellation in the odd sector while simultaneously exhibiting persistence or accumulation in the even sector.

Proposition 1. *Let a forcing protocol generate two response episodes with signed amplitudes $+u$ and $-u$, equal duration, and equal magnitude to leading order. If A is odd and Q is even under $u \mapsto -u$, then*

$$A(+u) + A(-u) = 0, \qquad Q(+u) + Q(-u) = 2Q(u) \geq 0. \tag{30}$$

Proof. Immediate from the defining parity relations. □

This conclusion is purely classical and depends only on parity class, not on any specific microscopic interpretation of the even scalar observable.

5 Response regimes

Let

$$K \equiv \kappa_{\mathrm{eff}}\, \tau_{\mathrm{eff}} \tag{31}$$

denote the effective gain-delay product. The reduced delayed-loop picture suggests four experimentally distinguishable regimes.

5.1 Regime I: one-pass delayed transport

For weak reinjection or no reinjection, the system remains in the transport regime. The key observation is a reproducible delayed arrival at the surface,

$$t_n^{\mathrm{det}} - t_n^{\mathrm{fire}} \approx \tau_h, \tag{32}$$

but no stable branch selection occurs.

5.2 Regime II: stable locking

At moderate effective gain, repeated event-triggered reinjection reinforces a preferred response cadence. Then one expects

$$t_{n+1}^{\mathrm{det}} - t_n^{\mathrm{det}} \to \text{constant}, \tag{33}$$

and the pulse-to-pulse response amplitude approaches a stable value

$$X_n \to X_\star. \tag{34}$$

5.3 Regime III: multistability and hysteresis

For larger K, multiple stable response branches may coexist. The experimentally visible signature is that two or more distinct stable operating states exist at the same control parameters, with branch occupancy depending on history, initial preparation, or perturbation.

5.4 Regime IV: irregular response

At still larger effective gain or under sufficiently strong nonlinear reinjection, stable locking can break down and the event-interval sequence becomes irregular:

$$\Delta t_n = t_{n+1}^{\mathrm{det}} - t_n^{\mathrm{det}} \not\to \text{constant}. \tag{35}$$

At this stage, the response may become broadband, intermittent, or chaotic in the practical experimental sense.

6 Experimentally accessible observables and protocol

A minimal experiment establishing the mechanism described above needs only the following measured quantities:

1. the one-pass hydrodynamic delay
$$\tau_h = t_{\rm det} - t_{\rm fire}, \tag{36}$$
2. the response amplitude sequence
$$X_n, \tag{37}$$
3. the event-interval sequence
$$\Delta t_n = t^{\rm det}_{n+1} - t^{\rm det}_n, \tag{38}$$
4. a signed angle-like response
$$A_n, \tag{39}$$
5. an even scalar cycle measure
$$Q_n \propto \int_{t_n}^{t_{n+1}} u^2(t)\,dt \quad \text{or} \quad Q_n \propto \int_{t_n}^{t_{n+1}} \omega_z^2(t)\,dt. \tag{40}$$

The minimal protocol is:

1. measure τ_h without reinjection,
2. verify enhancement of axial coherence under regular pulsing,
3. enable event-triggered reinjection,
4. sweep gain, latency, and background rotation,
5. map locking, multistability, and irregularity,
6. perform paired sign-reversal protocols and compare A with Q.

7 Discussion

The rotating cylinder with event-triggered reinjection is a classical delayed nonlinear loop. Its interest lies in the fact that the loop is not an abstract delay line, but a rotation-prepared fluid medium with finite transport time and a measurable axial response.

The rotating-fluid part provides the transport channel. The event-triggered reinjection provides the closure. The reduced delayed-loop model provides the branch-selection language. And the parity split provides a clean internal contrast between observables that reverse and observables that do not.

This makes the platform unusually economical. One and the same experiment can demonstrate finite transport delay, reinjection-based branch selection, multistability or irregularity, and the coexistence of cancelling odd observables with persistent even observables.

8 Conclusion

We have formulated a classical rotating-fluid platform in which axisymmetric forcing, finite axial transport time, and event-triggered reinjection combine to generate delayed-loop dynamics in a rapidly rotating fluid column.

The central structural result is

$$\text{rotation} + \text{finite transport time} + \text{reinjection} \Longrightarrow \text{delayed nonlinear loop behavior.} \tag{41}$$

Within this loop, parity separates the observable space into two distinct sectors:

$$A(-u_z) = -A(u_z), \qquad Q(-u_z) = Q(u_z). \tag{42}$$

Hence reversal-symmetric forcing can yield leading-order cancellation in a signed angle-like observable while preserving a nonzero scalar contribution in the even sector.

No quantum assumptions are required for any part of this argument. The result is entirely classical and rests only on rotating-fluid transport, event-defined reinjection, delay-induced selection structure, and parity.

A Compact reduced notation

For convenience, the main reduced parameters are

$$\tau_{\text{eff}} = \tau_h + \tau_{\text{det}} + \tau_{\text{ctrl}} + \tau_{\text{act}}, \tag{43}$$

$$K = \kappa_{\text{eff}}\,\tau_{\text{eff}}, \tag{44}$$

$$t_n^{\text{det}} = \inf\left\{ t > t_{n-1}^{\text{fire}} + t_{\text{ref}} : s(t) \geq s_{\text{th}} \right\}, \tag{45}$$

$$t_{n+1}^{\text{fire}} = t_n^{\text{det}} + \tau_d. \tag{46}$$

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